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Scheduling of Distributed LLM Serving on Heterogeneous GPUs

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```
$ make wordcount
gs -q -dSAFER -sDEVICE=txtwrite -o - \
    -dFirstPage=6 -dLastPage=11 report-submission.pdf | \
egrep '[A-Za-z]{3}' | wc -w
467
```

]

Declaration

I, Nathan Rignall of Churchill College, being a candidate for the Master of Philosophy in Advanced Computer Science, hereby declare that this report and the work described in it are my own work, unaided except as may be specified below, and that the report does not contain material that has already been used to any substantial extent for a comparable purpose.

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Abstract

Write a summary of the whole thing. Make sure it fits on one page.

Acknowledgements

This project would not have been possible without the wonderful support of . . . [optional]

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Chapter 1

Introduction

Chapter 2

Background

2.1 Transformers

Transformer models are a type of deep learning architecture specifically designed to handle sequential data such as natural language. Unlike recurrent neural networks (RNNs) which process sequences one step at a time, transformers can analyse entire sequences of data in parallel. This is achieved through self-attention, where each token in a sequence can weigh and incorporate information from all other tokens. Such attention is calculated by adding positional encodings to input embeddings which transforms each token into query, key and value vectors. Some model architectures also use multi-head attention, where multiple query, key and value vectors are calculated.

The first Transformer models were originally designed for encoder-decoder tasks such as translation, where an input sequence is encoded and decoded to an output sequence.

An encoder Processes input sequence Contextualized hidden representation No access to future tokens

Decoder Generates output tokens one-by-one Masked self-attention to prevent seeing future tokens

Encoder-only models are typically used for classification and sentence embeddings.

Decoder-only Used for autoregressive generation (language modelling, chatbots, coding assistants)

Transformers form the basis of modern natural language processing due to their scalability, parallelisation benefits, and excellent performance on long-range dependencies.

What is an LLM?

2.1.1 Computation Steps

There are several compute steps involved in a Transformer model's forward pass, during inference tasks. The process begins with Input Embedding and Positional Encoding where each token in the input sequence is converted into a vector, using a learned embedding matrix. Next, the model computes the Query (Q), Key (K) and Value (V) vectors for each token, this is calculated by projecting the input embeddings through learned linear transformations. Using the Q, K, and V values the model performs Scaled Dot-Product Attention

Models can also use Multi-Head Attention

Instead of computing a single attention operation, the model splits the Q, K, and V matrices into multiple "heads", concatenates the outputs

The Add and Normalisation steps follow attention - The original input is added to the output and normalised

Feed-Forward Network

Stacked Layers

Output

Diagram of Transformer architecture

2.1.2 Prefill and Decode

We can separate the processing phases of LLM model inference steps into two distinct categories, with clear differences in hardware processing requirements.

The prefill stage processes the input tokens to compute the transformer model's intermediate states. This includes calculating the positional embeddings and populating the KV cache, which are subsequently used to generate the first new token.

This specific phase is easily parallelized because at every stage of computation (per token) the full input sequence is known. Hence, this is viewed as a matrix-matrix operation which can easily saturate a GPU's available compute. *In reference to a transformer's computation steps, all stages are performed for every token in the sequence but can be heavily parallelised.*

The decode phase generates each output token one at a time, this is because each token computation is dependent on the previous token. Hence, each token must be calculated sequentially. The speed at which the decode phase can execute is heavily dependent on the speed at which weights, keys, values and activations can be loaded into the correct areas of GPU memory. Hence this phase usually saturates the memory bandwidth of a GPU before saturating the compute performance. *In reference to a transformer's computation*

steps, all stages are performed, but for only the new token generated - this is due key-value caching.

Diagram of Prefill vs Decode

2.1.3 Key Value caching

Key value caching is an optimisation used during inference steps for transformer-based decoder-only models such as GPT. This technique avoids recomputing the tensors for each input token for every decode token generated. Instead weights and key values are cached in GPU memory, to be used in the next iteration. At each decoding step the Q value of the new token is computed, reusing cached K and V values from earlier tokens in the sequence. This technique comes at the cost of using more GPU memory, reducing the maximum model size that theoretically could be loaded.

Diagram of Key Value cache

2.1.4 Mixture of Experts

A neural network architecture specifically design to scale up models

Instead of one big model Pool of expert networks For each input token, usually only a subset of these experts are activated Router chooses which experts to use based on the input

Designed to be compute efficient Only the activate experts consume compute Allows for scaling up a model's capacity (without linear increase in inference cost)

Good Big model cpacpaity Better scling lwas Felxiblke

Bad Routing instability Load balancing Complexity

2.1.5 DeepSeek Architecture

Text

2.2 Parallelism

Data parallelism refers to replicating a model across multiple processing units (GPUs/CPU/TPUs), with each replica receiving a different slice of the input data. Hence, this technique is generally used to increase throughput of an overall system, especially in batched inference scenarios

Example - batch of requests is split between $100 / 4 = 25$

Model parallelism involves partitioning the model itself across several processing units, where each device is responsible for a portion of the model's layers or operations. This is most helpful when a model cannot fit on a single device, due to memory demand greater than what is

Enables inference on devices that individually do not support

Every forward pass requires synchronisation, introducing a communication overhead that may negatively impact latency

Example

Tensor parallelism is a specialised form of model parallelism, generally used with large deep learning models.

Individual tensors are split across multiple devices (rather than whole layers)

Operator-Level parallelism Stuff

Expert Parallelism Mixture of experts

Expert parallelism occurs when different experts are placed on different devices and are activated in parallel for

Batching Stuff

2.3 Scheduler

Optimisation techniques...

2.3.1 Heuristic-Based Methods

- First-Come, First-Served (FCFS) - Round Robin (RR) - Min-Min / Max-Min - Priority-Based - Load Balancing (Best Fit, Worst Fit)

2.3.2 Metaheuristic Algorithms

- Genetic Algorithms (GA) - Particle Swarm Optimization (PSO) - Ant Colony Optimization (ACO) - Simulated Annealing (SA)

2.3.3 Search-Based

Search-based Optimisation refers to methods that systematically or semi-randomly explore a given search space to find an optimal solution. Such techniques are often used to tune a scheduler's parameters, but can also

Bayesian Optimisation (BO)

Random Search Sample search space uniformly at random + evaluate Context - randomly generate candidate schedules, and pick the best one found Search space is significant

Grid Search Brute force Discrete set of possible values for each parameter, then evaluates every combination Suffers from curse of dimensionality

2.3.4 Constraint Programming

Both Constraint Programming and Mathematical Optimisation approaches formulate scheduling activities as an optimisation problem, with formal mathematical constraints and objectives.

Linear Programming (LP) techniques can solve scheduler problems where the constraints and objectives can be modelled as only linear. All decision variables are continuous and can be used in scheduling algorithms when fractional assignments of work are valid.

Mixed Integer Linear Programming (MILP) solvers can consider integer variables such as binary assignment values, extending LP. In the context of scheduling, this enables decisions to be modelled using yes/no decisions and combinatorial choices.

In the case where constraints cannot be as linear, Non-Linear Programming (NLP) can model problems where either the objective and/or constraints are non-linear functions. Given problems are non-linear, it is harder to guarantee a globally optimal solution is generated.

Mixed Integer Programming (MIP)

Chapter 3

Related work

3.1 Case Studies

3.1.1 Thunderserve

Stuff

3.1.2 HexGen

Stuff

3.1.3 DistServe

Stuff

3.1.4 Splitwise

Stuff

3.1.5 Summary

Stuff (General trends)

3.2 Future

Stuff (Possible Improvements)

Chapter 4

Design and implementation

4.1 Simulator

In order to be able to develop an advanced heterogenous LLM scheduler, we need access to many high-performance GPUs to test our schedule configurations. Unfortunately, GPU-hours are prohibitively expensive and using such hours for “dummy” workloads to test a scheduler would be considered a waste of resources. Instead, we utilise the latest simulator technology to estimate the performance of LLM inference given a set of input parameters. Several simulators exist that attempt to model the performance of an LLM request by using estimation techniques.

Vidur (essentially uses linear regression)

Shallowsim is a specialist simulator designed to estimate the compute times for different LLM request types.

LLMCompass (Need to look up)

We Select Shallowsim for its transparency for calculations (simplicity). And support for Mixture of Expert models

4.1.1 Implementation

Prefill

Dense-MLA (dense Multi-head/Multi-query Attention compute)

TP-MLA (Dense-MLA plus the Tensor-Parallel all-reduce)

Dense-MLP (feed-forward network compute)

Shared-Expert (MoE expert run that every token visits) Routed-Expert (MoE expert run that only selected tokens visit)

Every token first passes through the Shared Expert, but a subset of tokens are also routed to extra specialists. This is modelled with the Routed Expert function.

Dispatch (All-to-All that ships tokens out to remote experts) Combine (All-to-All that brings tokens back)

Sending activations to the respective routed expert nodes is modelled using the Dispatch step, the Combine step then models how processed activations are collated back together. These two functions model the network overhead of LLM queries.

Compute Time - Dense MLA, Dense MLP (Dense Layers) - TP-MLA, Shared Expert, Routed Expert (Sparse Layers) Communication Timemake - Dispatch, Combine (Sparse Layers) Overlap Penalties - If Combine is longer than TP-MLA + Shared Expert - If Dispatch is longer than Routed Expert Sum - Compute + Communication + Any Overlap = Prefill Time

Decode

Load-KV Dense-MLA (as prefill) Sparse-MLA Dense-MLP (as prefill) Shared-Expert (as prefill) Routed-Expert (as prefill) Dispatch (as prefill) Combine (as prefill)

Compute Time - Dense MLA, Dense MLP (Dense Layers) - Sparse MLA, Shared Expert, Routed Expert (Sparse Layers) Communication Time - Dispatch, Combine (Sparse Layers) Overlap Penalties - TODO Sum - TODO = Per token decode time

4.1.2 Modifactions

What we have done to modify the shallowsim simulator.

As designed, shallowsim is build for profiling several different GPU configurations at once.

For our system, we are particularly interested in per-configuration results

4.1.3 Usage

Prefill Phase

Using the Shallowsim simulator we can measure the time t_{ms} (in milliseconds) to compute a single prefill request for a given set of parameters, and convert this to throughput R (in requests per second) via $R_{\text{prefill}} = \frac{1000}{t_{\text{ms}}}$.

Decode Phase

Shallowsim has support for calculating the time in milliseconds for a single token generation step. As we would typically expect several tokens to be generated, we cannot directly compute requests per second using Shallowsim, instead we must perform additional computations. The decode time of a single token is affected by the input/current sequence length, therefore we must measure the single token compute time for every token to be generated.

By measuring $t_{\text{ms},i}$ the compute time (in milliseconds) for the i th token, the total decode time for a request generating N tokens is $t_{\text{decode}} = \sum_{i=1}^N t_{\text{ms},i}$.

We can then convert this total time into throughput R_{decode} (in requests per second), using the same method as prefill. $R_{\text{decode}} = \frac{1000}{t_{\text{decode}}} = \frac{1000}{\sum_{i=1}^N t_{\text{ms},i}}$, where R_{decode} represents how many full decode requests per second the configuration can sustain.

4.2 Architecture

How the system will be constructed, how it is similar to existing work.

Like thunderserve

Separate into two distinct problems

Outer optimisation

Effectively an inner finite bin packing problem (statistical best solution)

Explain GPU islands... Scheduler direct traffic... Configure GPUs...

4.2.1 Inner Loop

Stuff

4.2.2 Outer Loop

Stuff

4.3 Workload Modeling

Statistical distributions (available from sources)

4.3.1 Input Sequence Length

To achieve more fine-grained control of request scheduling, dependant on input sequence, we model the input request length as a probability distribution. Hence, for a chosen workload type, there is a specific probability of request arriving with given sequence length.

Let $w \in W$ denote a workload type from the set of all workload types W . For each workload type w , we define a probability mass function $p(s \mid w)$, which represents the probability of a request having sequence length $s \in S$ under workload w . This distribution satisfies:

$$\sum_{s \in S} p(s \mid w) = 1 \quad \text{for all } w \in W$$

Instead of only considering the distribution of sequence lengths during assignment, to model system throughput, we use them to find the most optimal pairing of sequence length to GPU island. Therefore, specific GPU islands can be configured to most optimally perform prefill operations for a specific sequence length. When a request arrives, we can direct it to the GPU islands targeted to serve that request length, defined in the schedule.

4.3.2 Decode Length

While decode length can also be modelled using a probability density function, this information (specific distribution) is less useful from a scheduling perspective. In contrast to input sequence length, the decode length is not known the moment a request arrives. Instead, the generation process proceeds token-by-token until an explicit or implicit stop condition is met (e.g. end-of-sequence token).

We can assume that a decode request must complete on the GPU island it started processing on, as switching island would incur additional communication overhead and significantly impact the request response time.

For some workloads there is a small correlation between input sequence length and decode length, but this relationship is often not significant. Therefore, at the point of request arrival we do not have enough information about the decode length to make informed per-decode length scheduling decisions.

Hence, creating specific GPU configurations for short, medium, long, etc. decode lengths would likely be inefficient. As we cannot accurately predict decode length, there is a high

probability that a decode request would have to move island.

Prior work has demonstrated that it can be possible to predict decode length using the input text contents.

Instead, we model our scheduler using an average decode length for a given workload type and design every GPU decode island to support the maximum decode length expected. Unlike the sequence length, the decode length is simply being used for system modelling activities to characterise performance, not direct traffic.

However, we can still make scheduling decisions for the decode phase using the sequence length as a feature, given the maximum decode length affected by the input length. This is due to all the input and decode tokens KV values being stored in GPU memory.

4.4 Objectives

4.4.1 Throughput

Demonstrated in subsection 4.1.3, the Shallowsim simulator has support for estimating the throughput of a single request of a static input sequence and decode length. It is highly improbable that an entire workload consists of a single request type, instead we expect a distribution of sequence lengths, as shown in section 4.3. Hence, we need a method to accurately calculate the throughput of a system, with a variety of input sequence.

Using the probability distribution of the workload, we can compute the weighted average of throughput, using measured performance values for each sequence length. Let $R_{\text{prefill}}(s)$ denote the measured prefill throughput (in requests per second) for a request of sequence length s . The expected prefill throughput for workload w , denoted $\mathbb{E}[R_{\text{prefill}} | w]$ (expectation), is then given by:

$$\mathbb{E}[R_{\text{prefill}} | w] = \sum_{s \in S} R_{\text{prefill}}(s) \cdot p(s | w)$$

In addition to prefill, we can also compute the expected throughput for the decode phase. While prefill throughput depends solely on the input sequence length s , decode throughput typically depends on both the input length s and the output (decode) length d . For simplicity, we assume a fixed average decode length d as a parameter. Let $R_{\text{decode}}(s, d)$ denote the measured decode throughput (in requests per second) for an input sequence length s and average decode length d . Given a workload-specific sequence length distribution $p(s | w)$, the expected decode throughput for workload w is:

$$\mathbb{E}[R_{\text{decode}} | w, d] = \sum_{s \in S} R_{\text{decode}}(s, d) \cdot p(s | w)$$

To account for the complete request processing pipeline, we define the overall expected throughput as the minimum of the expected prefill and decode throughputs. Since requests cannot be served faster than the slowest stage, the effective throughput is bounded by the bottleneck. Let $\mathbb{E}[R_{\text{prefill}} | w]$ denote the expected prefill throughput for workload w , and $\mathbb{E}[R_{\text{decode}} | w, d]$ the expected decode throughput for workload w and average decode length d . The overall expected throughput is given by:

$$\mathbb{E}[R_{\text{overall}} | w, d] = \min(\mathbb{E}[R_{\text{prefill}} | w], \mathbb{E}[R_{\text{decode}} | w, d])$$

This value represents the effective throughput achievable by the system under the given workload and output length assumptions.

4.4.2 Percent Allocation

Previous calculations assumed that a singular GPU island is contributing to the overall throughput of the system. In reality, we would like to be able to construct several GPU islands and assign each island to a request type it is best suited for. Hence, we allocate a percentage of GPU resources for a given island to a given sequence length it specialises in. Let I be the set of all GPU islands in the system, which we partition into two separate subsets I_{prefill} and I_{decode} . Let $x_i(s)$ denote the fraction of overall GPU capacity on island i devoted to handling requests of length s . We normalise these allocations so that

$$\sum_{s \in S} x_i(s) = 1 \quad \text{for every } i \in I.$$

By including $x_i(s)$ into our previous per-length throughput models, the expected prefill throughput under workload w becomes

$$\mathbb{E}[R_{\text{prefill}} \mid w] = \sum_{i \in I_{\text{prefill}}} \sum_{s \in S} x_i(s) R_{\text{prefill}}^{(i)}(s) p(s \mid w),$$

while the expected decode throughput for average decode length d is

$$\mathbb{E}[R_{\text{decode}} \mid w, d] = \sum_{i \in I_{\text{decode}}} \sum_{s \in S} x_i(s) R_{\text{decode}}^{(i)}(s, d) p(s \mid w).$$

Since end-to-end request processing cannot exceed its slowest stage, the overall expected throughput is simply the bottleneck of these two values:

$$\mathbb{E}[R_{\text{overall}} \mid w, d] = \min(\mathbb{E}[R_{\text{prefill}} \mid w], \mathbb{E}[R_{\text{decode}} \mid w, d]).$$

4.5 Constraint Optimisation

We can combine the previous throughput calculations to form a simple constraint optimisation problem with an objective to maximise the total expected throughput R and minimise the deviation Δ from the target workload distribution $p(s | w)$.

4.5.1 Non-Linear Problem

The following statement describes a sample non-linear optimisation problem that can be solved for either prefill or decode throughput.

Input Data

- I (set of GPU islands),
- S (set of sequence lengths),
- $b_i(s)$ (benchmark throughput of island i on length s),
- $p(s | w)$ (target probability of length s under workload w), $\sum_{s \in S} p(s | w) = 1$,
- λ (trade-off parameter, $\lambda > 0$, balancing throughput vs. deviation).

Decision Variables

$$x_i(s) \geq 0 \quad (\text{fraction of island } i\text{'s GPU capacity devoted to length } s).$$

Constraints

$$\sum_{s \in S} x_i(s) = 1 \quad \forall i \in I.$$

Intermediate Quantities

1. Performance per cell:

$$r_i(s) = b_i(s) x_i(s).$$

2. Total performance:

$$R = \sum_{i \in I} \sum_{s \in S} r_i(s) = \sum_{i \in I} \sum_{s \in S} b_i(s) x_i(s).$$

3. Performance by sequence length:

$$R_s = \sum_{i \in I} r_i(s) = \sum_{i \in I} b_i(s) x_i(s), \quad \forall s \in S.$$

4. Achieved share:

$$\sigma_s = \frac{R_s}{R}.$$

5. Deviation from target:

$$\delta_s = \left| \sigma_s - p(s \mid w) \right|, \quad \Delta = \sum_{s \in S} \delta_s.$$

Objective

Maximise

$$R - \lambda \Delta,$$

where

$$R = \sum_{i \in I} \sum_{s \in S} b_i(s) x_i(s), \quad \Delta = \sum_{s \in S} \left| \sigma_s - p(s \mid w) \right|, \quad \sigma_s = \frac{\sum_{i \in I} b_i(s) x_i(s)}{R}.$$

4.6 Scheduling Assumptions

We can make several assertions and assumptions in an attempt to reduce the search space of the scheduler generation process. These effectively apply heuristics to our problem, that are well crafted to simplify the computation requirements while retaining solution quality.

4.6.1 Resolution

We can use the simulator to benchmark the prefill and decode throughput of an island for a given input sequence length. While it is feasible to consider each sequence length individually, we can use a binning approach to reduce the search space of the scheduler. The width of the bins effectively controls the resolution of the solution generated, which can be dynamically adjusted depending on input parameters.

4.6.2 Optimal Parallelism

To further reduce the search space of the scheduler, we can assume for a given model and GPU type, there exists an optimal parallelism configuration that is consistent across input sequence lengths. Demonstrated in Figure 4.1 we can see that for all GPUs, $tp = 64$ and $dp = 1$ is the fastest parallelism configuration for all sequence lengths above 128 tokens. Given we expect the majority of input requests to be over 128 tokens, we can reason that there is an *optimal* GPU prefill configuration.

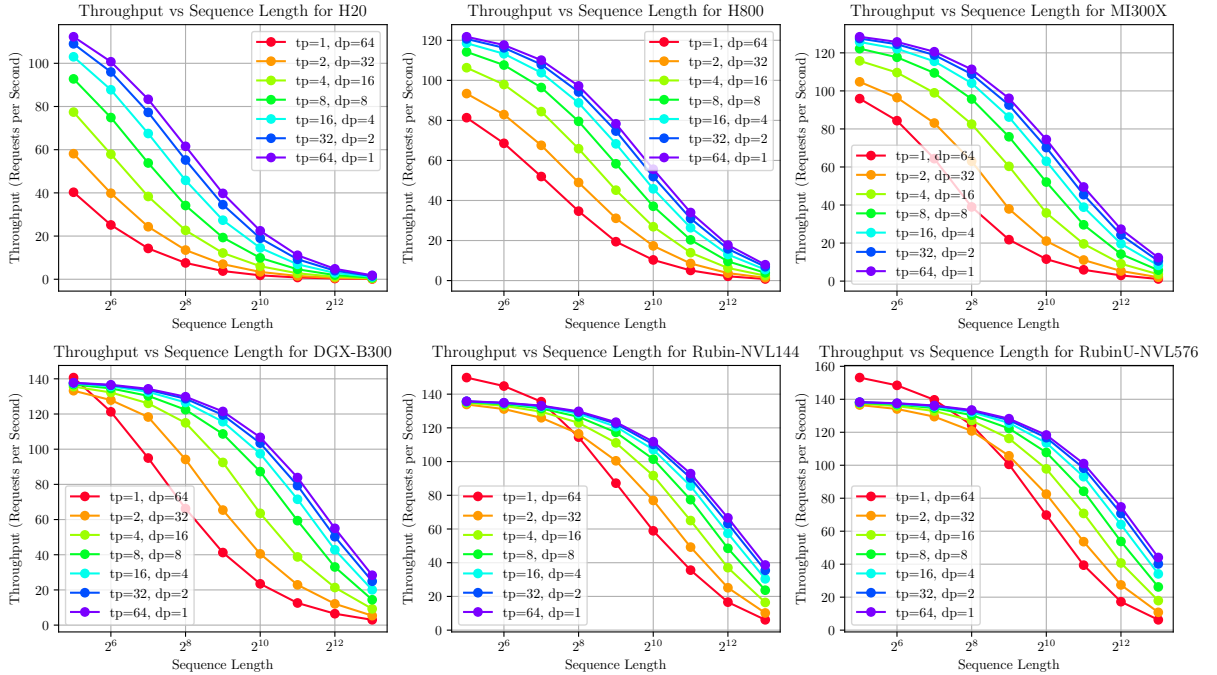


Figure 4.1: Stuff

By pre-computing this optimal configuration, we significantly reduce the number of parameters considered by the scheduler. This is because an island cannot be formed with sev-

eral different parallelism configurations simultaneously. And hence, the scheduler would have to contain a variable for each sequence-length-parallelism combination.

4.7 Inner Solver

Treating the

4.8 Outer Optimiser

Stuff

Chapter 5

Evaluation

Chapter 6

Summary and conclusions

Appendix A

Technical details, proofs, etc.

Appendices are for optional materials that is not essential to understanding the work, and that the examiners are not expected to read, but that will be of value to readers interested in additional, in-depth technical detail.

A.1 Lorem ipsum

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.